CHRONIC

CHRONIC

OVERDRIVE

CHRONIC

CHRONIC

NTWK: NEURO.INTL

CHAN: 02

SIGNAL: CLEAR

STATUS: STABLE

TABLE OF CONTENTS

SYNOPSIS

In this issue of NeuroSphere we dive deep into the adversities of stress and burnout. We explore the physiological realities of neuroinflammation and the 'amygdala hijack' that alters everyday logic as we dissect the misconception of burnout simply being a lack of willpower. Inside, you'll uncover the science behind how stress impairs our working memory, how emotions act as airborne pathogens through secondhand stress and where you land on the pressure scale with our interactive 30-Second Quiz!

EDITOR'S NOTE

Dear Readers,

Burnout is classified as a mental slump or a simple crisis of willpower and is frequently worn as a badge of honor. Neuroscientific research suggests otherwise by studies proving that chronic stress plays a role in remodeling the human nervous system.
In this month's edition titled ‘Chronic Overdrive: The Brain Under Pressure’ we look at what happens when our minds are pushed past their biological limits.

Your brain experiences a physical takeover when you are constantly under pressure. Chronic cortisol exposure weakens the prefrontal cortex (executive control center) and control is lent to the amygdala (fear center). This internal shift hinders our logical development thereby forcing us to default to short-sighted habits. We also explore the physical reality of burnout as a state of neuroinflammation where the brain and body become resistant to their own stress-regulating hormones.

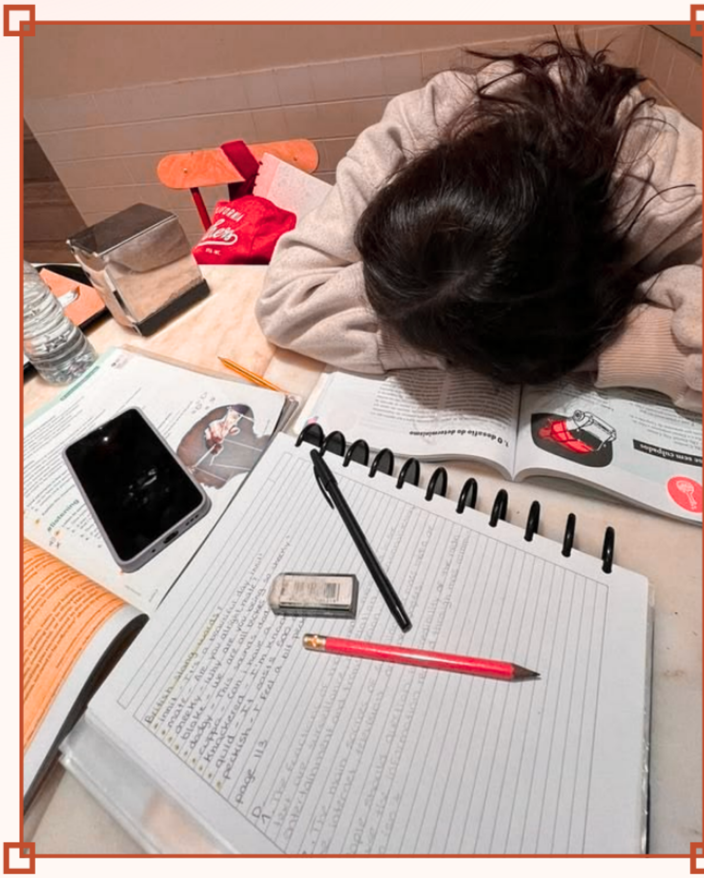
To conclude, we look at how stress acts as an airborne pathogen. Mirror neurons and subconscious micro-expressions absorb the neurological strain of the people around us shifting the collective intelligence of entire rooms.
As you flip through this issue take a look at your nervous state. Are you managing your pressure successfully or is your brain running on chronic overdrive?

Warmly,
The Editor
Amna Ashker

REDEFINING BURNOUT

REDEFINING BURNOUT AS A PHYSICAL CONDITION OF THE NERVOUS SYSTEM RATHER THAN A MENTAL WEAKNESS OR LACK OF MOTIVATION

The fast-paced world of modern society is characterized by constant productivity and the chronic stress that accompanies it. While many people struggle to sustain the demands of workplace environments, those experiencing persistent exhaustion are frequently told to simply “try harder” or “push through” this phenomenon known as burnout.
Burnout is defined as “a syndrome conceptualized as resulting from chronic workplace stress that has not been successfully managed” (World Health Organization, 2019).



Burnout is defined as “a syndrome conceptualized as resulting from chronic workplace stress that has not been successfully managed” (World Health Organization, 2019).

Despite its prevalence, burnout is often dismissed as a sign of weakness, poor time management, or a lack of resilience. However, growing evidence from neuroscience suggests that burnout is not merely a psychological state, but a condition associated with measurable physiological changes in the brain and body. As research of burnout continues to evolve, redefining it as more than a personal failing is becoming an increasingly important issue to understand and address.

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THE SCIENCE BEHIND BURNOUT

While burnout is often mistaken for everyday stress, it extends beyond this. When stress persists over time, the sympathetic nervous system's fight-or-flight response and hypothalamic-pituitary-adrenal (HPA) axis remain repeatedly activated, leading to sustained release of cortisol and other stress hormones to keep the body alert. The parasympathetic nervous system is then designed to return the body to baseline after a stressor is resolved (Chu et al., 2024; Harvard Health, 2011). However, under intense workplace or environmental demands, this system can become dysregulated and unable to fully shut down.

Over time, this activation is what contributes to physiological and neurological changes associated with burnout as the final stage of exhaustion and depletion. From a scientific viewpoint, studies have found brain function to be largely affected. The functioning of the prefrontal cortex that is responsible for executive function, decision-making, and emotional regulation is impaired– causing gray matter connections of synapses and local neuron networks to be lost (Arnsten & Shanafelt, 2021). Meanwhile, the amygdala becomes enlarged and increases its functional activity (Khammissa et al., 2022). This means that even minor inconveniences can trigger the same fight-or-flight responses as physical danger.

ARE YOU RUNNING ON EMPTY?

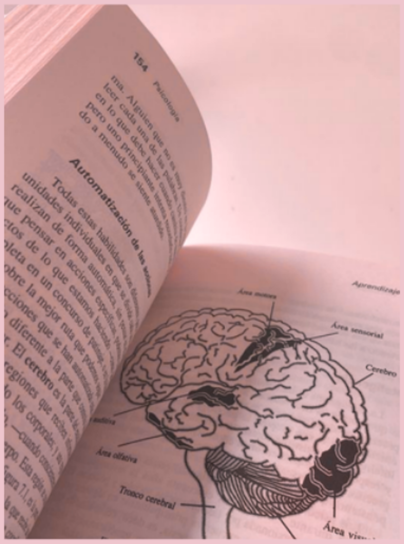
KEY TERMS

HPA Axis: Body's central stress-response system that triggers cortisol release.

Cortisol: Stress hormone that increases alertness but can become harmful when chronically elevated.

Prefrontal Cortex (PFC): Controls decision-making, focus, and emotional regulation; prolonged stress weakens its function.

Amygdala: Processes fear and threats; chronic stress can make it more reactive.



WHEN BURNOUT BECOMES PHYSICAL

Moving beyond the brain, burnout grounds itself in reality through three dimensions, namely: energy depletion or exhaustion, increased mental distance or cynicism related to one's job, and reduced professional efficacy (World Health Organization, 2019). For individuals suffering from burnout, the effects are extensive and can include irritability, sleep impairment, difficulties in cognitive and executive functioning, feelings of helplessness, dissatisfaction with personal accomplishments, and ineffective coping (Khammissa et al., 2022). While these symptoms may seem psychological on the surface, they have tangible consequences in the workplace that are often mistaken for laziness. Not only can it reduce productivity and increase the likelihood of errors, it can also make it difficult to remain engaged with coworkers or clients. Tasks that were once manageable can begin to feel overwhelming, creating a cycle in which chronic stress reaffirms exhaustion and worsened performance.

CONCLUSION

While burnout may seem easy to dismiss, it is more than the occasional lack of motivation experienced from time to time. Modern society often falls into the trap of seeking to constantly produce the next output; however, the chronic stress and burnout from this lifestyle is not without consequence– altering and impairing how the brain functions. Recognizing burnout's neurological and physiological basis is the first step towards addressing it more effectively. Understanding its effects allows for a more accurate and compassionate perspective, especially given that it is generally viewed in a negative light. Redefining burnout, then, becomes not only a scientific necessity, but a societal one.

REFERENCES

1. World Health Organization. (2019, May 28). Burn-out an "occupational phenomenon": International classification of diseases. https://www.who.int/news/item/28-05-2019-burn-out-an-occupational-phenomenon-international-classification-of-diseases
2. Chu, B., Marwaha, K., Ayers, D., & Sanvictores, T. (2024, May 7). Physiology, Stress Reaction. PubMed; StatPearls Publishing. https://www.ncbi.nlm.nih.gov/books/NBK541120/
3. Understanding the stress response - Harvard Health. (2011, June 15). Harvard Health. https://www.health.harvard.edu/healthy-aging-and-longevity/understanding-the-stress-response
4. Arnsten, A. F. T., & Shanafelt, T. (2021). Physician Distress and Burnout: The Neurobiological Perspective. Mayo Clinic Proceedings, 96(3), 763–769. https://doi.org/10.1016/j.mayocp.2020.12.027
5. Khammissa, R. A. G., Nemutandani, S., Feller, G., Lemmer, J., & Feller, L. (2022). Burnout phenomenon: neurophysiological factors, clinical features, and aspects of management. Journal of International Medical Research, 50(9), 030006052211064. https://doi.org/10.1177/03000605221106428

THE BRAIN UNDER PRESSURE

THE BRAIN UNDER PRESSURE AND WHY STRESS MAKES THINKING HARDER

The experience of blanking out on a test or forgetting the answer to a simple question is not uncommon for many. In moments of high pressure, information that once seemed easy to recall can suddenly feel out of reach. While stress is thought to sharpen focus and improve performance, it can also interfere with the brain's ability to think clearly. Rather than simply feeling overwhelming, stress can alter how the brain functions making everything harder to think through.

×

What if forgetting the answer isn't a sign that you didn't learn it, but that your brain thinks you're in danger?

THE THINKING BRAIN AND STRESS

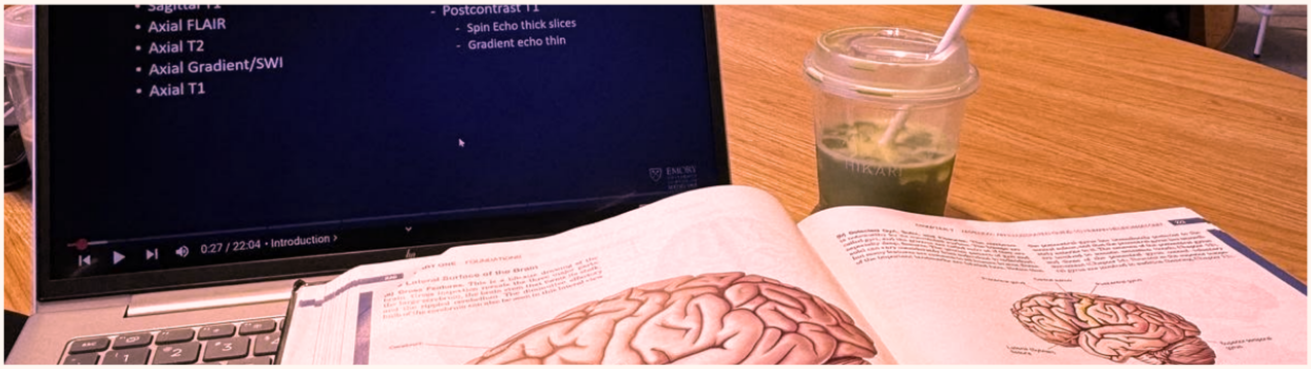
Short-term stress, or the body's temporary fight-or-flight response to an immediate threat, is established as a beneficial survival mechanism, enhancing immune function as well as mental and physical performance (Dhabhar, 2018). This helps explain why deadlines can increase productivity and moderate levels of test anxiety can in turn help improve focus. However, when stress becomes too intense in a given moment, cognitive processing begins to shift. In high-pressure situations, it becomes much harder to think as the brain goes into survival mode: here, the brain shifts resources away from higher-order thinking in the prefrontal cortex and toward the threat detection system of the amygdala that overrides complex thinking processes (Arnsten et al., 2015).

When this stress response remains partially active and becomes chronic due to continuous stressful situations, it not only causes temporary performance changes but can also begin to alter brain structure. In the long run, this can lead to lasting negative effects on cognition. Research suggests that prolonged exposure to stress hormones can reduce synaptic connectivity and contribute to shrinkage in the hippocampus, impairing working memory efficiency (Kumar et al., 2013; Carvalho et al., 2025). This is what allows individuals to hold and manipulate information – making it more difficult to organize thoughts, retain information, and keep track of multiple ideas at once when under pressure. It additionally weakens the prefrontal cortex's synaptic connectivity and persistent firing while simultaneously strengthening amygdala hyperactivity, greatly weakening cognitive abilities responsible for thought (Woo et al., 2021; Zhang et al., 2018).

REAL LIFE IMPLICATIONS

While stress doesn't reduce intelligence, it actively reduces access to cognitive resources that translate into measurable difficulties in everyday functioning. For students, this can look like blanking out during exams or struggling to retrieve information that was previously well-learned. When working memory becomes overloaded under stress, it can also lead to overthinking where anxiety further disrupts performance.

The effects extend to the workplace as well. Under pressure, decision-making requires more effort, often leading to decision fatigue throughout the day. Tasks that require problem-solving, planning, or clear communication can become more difficult, and individuals may be more prone to errors when working under tight deadlines or high-stakes situations



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REFERENCES

1. Dhabhar F. S. (2018). The short-term stress response - Mother nature's mechanism for enhancing protection and performance under conditions of threat, challenge, and opportunity. Frontiers in neuroendocrinology, 49, 175–192. https://doi.org/10.1016/j.yfrne.2018.03.004

CONCLUSION

The effects of stress on the brain are significant– having impacts on how individuals are able to think through complex problems. While small amounts of stress can be healthy, the chronic stress that appears in modern environments reveals how performance is not only effort-based, but also dependent on whether the brain is able to function under the right conditions. Ultimately, the ability to think clearly occurs when the brain is operating under conditions that support reasoning rather than survival.

1. Arnsten, A. F. T., Raskind, M. A., Taylor, F. B., & Connor, D. F. (2015). The effects of stress exposure on prefrontal cortex: Translating basic research into successful treatments for post-traumatic stress disorder. Neurobiology of Stress, 1(1), 89–99. https://doi.org/10.1016/j.ynstr.2014.10.002
2. Kumar, A., Rinwa, P., Kaur, G., & Machawal, L. (2013). Stress: Neurobiology, consequences and management. Journal of pharmacy & bioallied sciences, 5(2), 91–97. https://doi.org/10.4103/0975-7406.111818
3. Carvalho, C., Reis, C., & Serrano, M. (2025). Exploring the Link Between Stress and Working Memory in Adults. International journal of environmental research and public health, 22(12), 1773. https://doi.org/10.3390/ijerph22121773
4. Woo, E., Sansing, L. H., Arnsten, A. F. T., & Datta, D. (2021). Chronic Stress Weakens Connectivity in the Prefrontal Cortex: Architectural and Molecular Changes. Chronic stress (Thousand Oaks, Calif.), 5, 24705470211029254. https://doi.org/10.1177/24705470211029254
5. Zhang, X., Ge, T. T., Yin, G., Cui, R., Zhao, G., & Yang, W. (2018). Stress-Induced Functional Alterations in Amygdala: Implications for Neuropsychiatric Diseases. Frontiers in neuroscience, 12, 367. https://doi.org/10.3389/fnins.2018.00367

30-SECOND QUIZ

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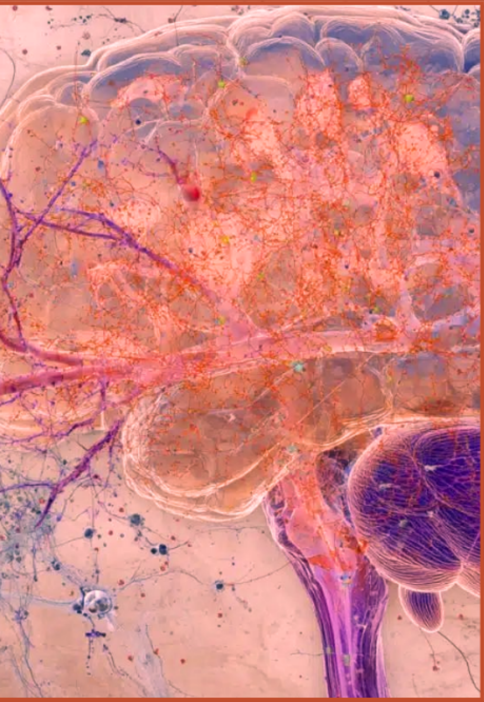
Tally how many of these statements describe you over the past two weeks:

- [ ] **Decision Fatigue:** I am overwhelmed by small choices. I typically choose whatever is easiest.
- [ ] **Brain Fog:** My mind feels scattered. I switch between tasks without finishing them.
- [ ] **High Reactivity:** Small setbacks or interruptions feel like major emergencies.
- [ ] **Stress Absorbency:** I instantly catch and feel the tense energy of the people around me.

Your results:

- **0–1 Statements: Prefrontal Dominance (Optimal)**
  - Your brain's executive center is firmly in control and so your focus is sharp and emotional reactivity low.
- **2–3 Statements: Allostatic Strain (The Tipping Point)**
  - Your nervous system is accumulating wear and tear. Your brain is running low on fuel and lets habits loop and mental fog creep in.
- **4 Statements: Chronic Overdrive (The Hijack Zone)**
  - Your amygdala has taken the keys. Your brain's ‘smoke detector’ becomes hyper-sensitive and treats daily tasks as survival threats.



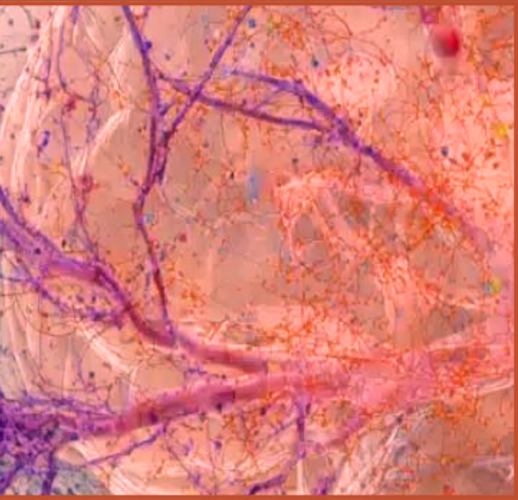
THE AMYGDALA HIJACK

STRESS IN THE MODERN WORLD

The feeling of stress is highly prevalent within society daily, from deadlines to academic pressure, work to financial concerns, etc. Stress is necessary in moderate levels to increase awareness and coping mechanisms when faced with challenges. While occasional stress may have some positive effects, long-term exposure is harmful. When stress becomes chronic, it isn't just temporary discomfort, stress hormones can alter the brain itself.

THE BRAIN'S BALANCE: LOGIC AND EMOTION

Primary brain regions affected by prolonged exposure to stress hormones are the prefrontal cortex and amygdala, both responsible for different cognitive processes while also often working together. The prefrontal cortex, located at the front of the brain, controls planning, decision-making, impulse control, and emotional regulation. The amygdala, however, is most commonly known for processing emotions like fear and potential threats. Normally, these regions collaborate and allow people to react to situations properly. The amygdala recognizes emotional stimulation, while the prefrontal cortex assesses the situation and decides what action to take about it.



BUILT FOR SURVIVAL, NOT CONSTANT STRESS

During times of stress, the body produces cortisol and adrenaline which make the body ready to face danger. This adaptation was developed through evolution as a mechanism to allow humans to defend themselves from potential threats in their environment. Although the reaction proves to be useful in cases of emergency, it was never intended to be kept on permanently. As a result, if the body stays exposed to high levels of cortisol for extended periods of time, it will lead to changes within the brain which would eventually impact its cognitive capabilities (McEwen & Morrison, 2013).

WHEN STRESS REWIRES THE BRAIN

According to research, chronic stress decreases the activity of the prefrontal cortex but increases the activity of the amygdala (Arnsten, 2009). Evidence indicates that excessive stress results in decreased connectivity between different networks in charge of executive control functions such as attention, emotion regulation, and strategic thinking (Cardoner et al., 2023). In turn, the amygdala becomes more sensitive to stress, resulting in increased perception of potential threats (Cardoner et al., 2023).

This change is commonly described as an "amygdala hijack." In these situations, people's emotions can overshadow their logic. Instead of rationally analyzing facts, people might be impulsive, defensive, or concentrate on pressing matters only. In other words, the brain chooses survival mode and disregards long-term planning.

REFERENCES

1. Arnsten, A. F. T. (2009). *Stress signalling pathways that impair prefrontal cortex structure and function.* Nature Reviews Neuroscience, 10(6), 410–422.
2. Cardoner, N., et al. (2023). *Impact of Stress on Brain Morphology: Insights into Structural Changes in the Amygdala, Hippocampus, and Prefrontal Cortex.*
3. Liston, C., McEwen, B. S., & Casey, B. J. (2009). *Psychosocial stress reversibly disrupts prefrontal processing and attentional control*. PNAS, 106(3), 912–917.
4. McEwen, B. S., & Morrison, J. H. (2013). *The brain on stress: Vulnerability and plasticity of the prefrontal cortex over the life course*. Neuron, 79(1), 16–29.
5. Sun, S., et al. (2023). *Functional connectivity between the amygdala and prefrontal cortex.*

THE REAL WORLD CONSEQUENCES

There is a wide range of ways in which this change manifests itself. Stressful students tend to have poor concentration and memory. Professionals working in high-stress jobs will have difficulty making complex decisions. Moreover, studies show that chronic stress limits one's cognitive flexibility, preventing adaptation to changes and taking into consideration various perspectives (Liston et al., 2009).

CAN THE BRAIN RECOVER?

Despite all this, the brain retains plasticity all throughout life - neuroplasticity. There have been findings that indicate that sufficient sleep, exercise, support systems, and interventions involving mindfulness can improve prefrontal cortex functionality and minimize amygdala overactivity. Although chronic stress may cause a change in the brain, most of the changes could be reversible through coping mechanisms.

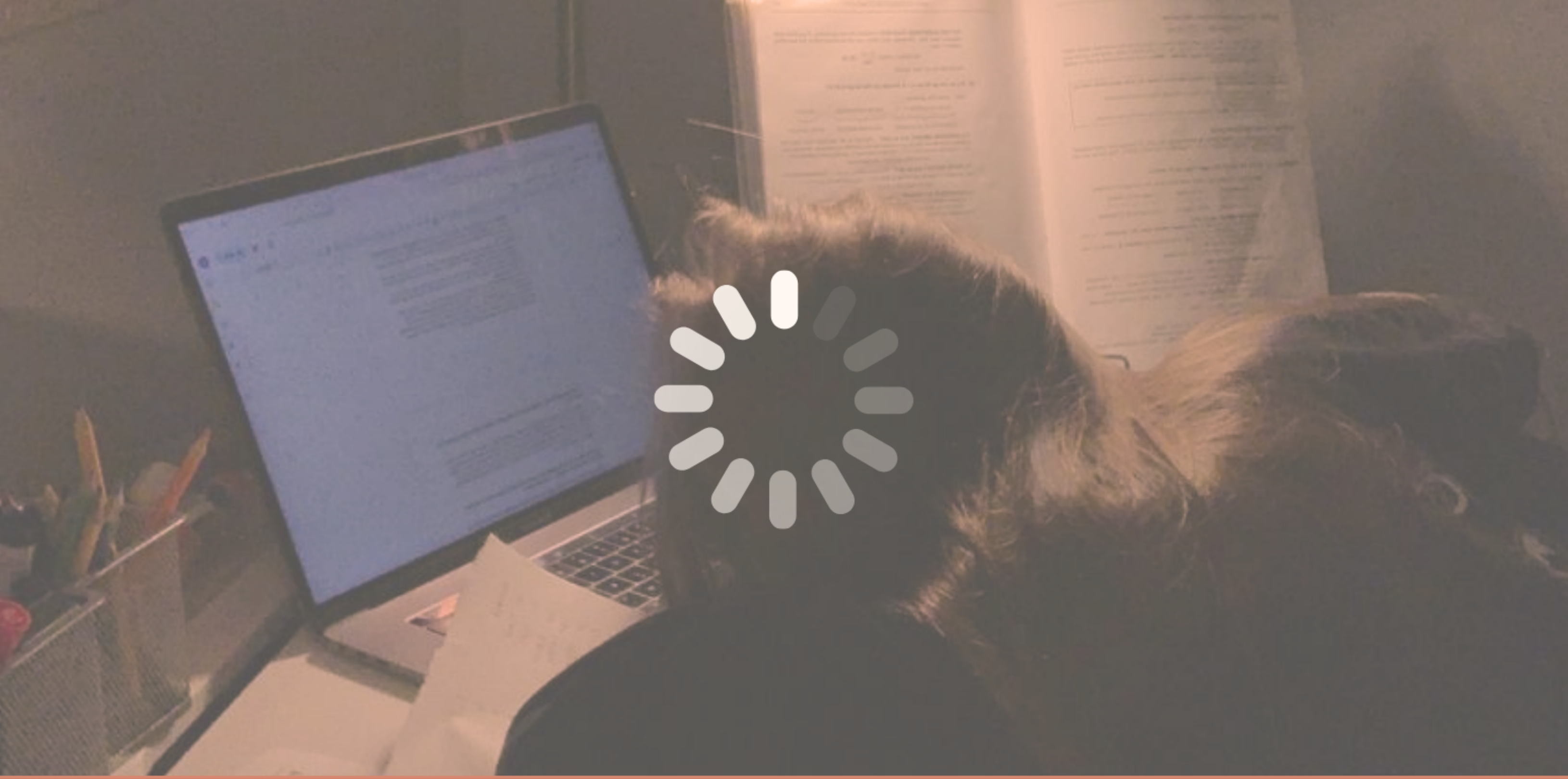
In this age where time is fast-moving and demanding on humans, it is becoming vital to understand the impact of stress on the brain. Stress does not only interfere with one's mood and productivity. Stress can actually rewire the brain, changing the balance of control from the part of the brain concerned with reasoning to the part of the brain meant for survival.

WHO'S DRIVING YOUR BRAIN?

Scenario: You receive a bad grade on a test, *what do you do?*

- [ ] Immediately panic and think the worst
- [ ] Take a breath and make a plan

Panic → Amygdala (Survival Mode)
Plan → Prefrontal Cortex (Thinking Mode)



THE NEUROSCIENCE OF SECONDHAND STRESS

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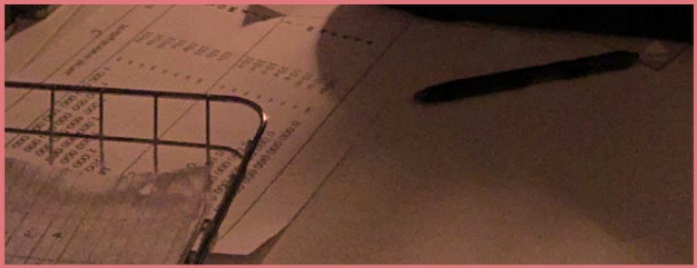
STRESS DOESNT STAY WITH ONE PERSON

Stress is often viewed as a personal experience. We associate it with our own responsibilities, deadlines, and challenges. However, growing evidence suggests that stress does not always remain confined to the individual experiencing it. Through social interactions, emotional cues, and physiological responses, stress can spread from one person to another. This phenomenon, commonly known as stress contagion or secondhand stress, demonstrates the powerful influence that social environments have on the brain.

Humans are very social creatures. In the course of evolution, survival became closely related to being able to detect any signs of potential danger in the surrounding people. If there were even the smallest signs that one of the members of the community detected something dangerous, then immediate reaction to that person's fear would increase the likelihood of surviving. Consequently, the brain of humans developed a mechanism which reacts keenly to the emotional state of other people.



THE SCIENCE OF EMOTIONAL CONTAGION

This can be done through the process of emotional contagion, whereby an individual unconsciously mimics and assimilates the emotions of other people within their environment. Facial expressions, tone of voice, posture, among other social cues, serve as continual sources of information about how others feel. The brain automatically receives and interprets such social information without even realizing it. As it turns out, seeing someone else go through a stressful situation activates the same neural circuitry in an observer as the stress experienced personally (Hatfield et al., 1993).

It has been proven that simply watching someone go through a stressful situation can cause physiological changes in the observer. In one well-known experiment, individuals who watched a person do something stressful had elevated levels of cortisol even though they did not experience the stressor (Engert et al., 2014).

THE BRAIN’S ALARM SYSTEM

The amygdala is one of the key parts of the brain in charge of second-hand stress. The amygdala constantly scans the surroundings looking for possible dangers and reacts not only to immediate threats but also to social stress signals. If someone around looks nervous or frightened, the amygdala may perceive these signs as a signal about the presence of danger and start to produce stress hormones.

STRESS IN EVERYDAY LIFE

Second-hand stress is very common in daily lives. Parental stress leads to higher stress reactions in children at home. Students become stressed during exams because of their classmates. There might be the phenomenon of second-hand stress at workplaces, as well. Stress contagion in the workplace happens when there is high-level stress perception communicated by the leaders or co-workers (Westman, 2001).

Ironically, the very neural mechanisms which enable stress contagion are also linked to empathy and bonding. The process of recognizing and reacting to the emotions of other people is crucial for cooperation, nurturing, and building relationships. In such a context, one might say that secondhand stress can be considered a side effect of the brain's amazing ability to empathize.

PROTECTING OURSELVES FROM STRESS

It should be mentioned that exposure to the stress of other people on a regular basis can be detrimental to mental health. The staff of healthcare institutions, nurses, therapists, and other workers regularly have to deal with stressful or even traumatizing situations. Therefore, knowing more about secondhand stress from the perspective of neuroscience can be helpful in dealing with the problem.

According to recent studies, social support can serve as a protection against many of these consequences. Social connections, supportive environment and positive culture in the workplace can alleviate stress reactions and increase resilience (Peen et al., 2021). In the same way stress can spread via social networks, serenity, trust and emotional support can affect our wellbeing through others.

YOUR SUPPORT CIRCLE

Who helps you feel calmer when you're stressed?

____

____

____

FINISH THE SENTENCE

When I am stressed it helps when someone...

- ☐ Listens
- ☐ Gives Advice
- ☐ Encourages
- ☐ Other

Talk it **OUT**, *Don't carry it* **ALONE**

MENTAL HEALTH IS COLLECTIVE

In today's connected world, the consequences of stress do not stop at one person. Through communication and relationships, through personal interactions and social environment, emotions can spread from one person to another, affecting not only psychological state but also physiological one. The understanding of the impact of secondary stress shows one thing very clearly: mental health is something to protect on a collective level.

REFERENCES

1. Engert, V., Plessow, F., Miller, R., Kirschbaum, C., & Singer, T. (2014). *Cortisol increase in empathic stress is modulated by emotional closeness and observation modality*. Psychoneuroendocrinology, 45, 192–201.
2. Hatfield, E., Cacioppo, J. T., & Rapson, R. L. (1993). *Emotional contagion*. Current Directions in Psychological Science, 2(3), 96–100.
3. Peen, N. F., et al. (2021). *Convergent neuroendocrine mechanisms of social buffering and stress contagion*.
4. Westman, M. (2001). *Stress and strain crossover*. Human Relations, 54(6), 717–751.
5. Richer, M. J., et al. (2024). *The contribution of vulnerability to emotional contagion to stress resilience*.

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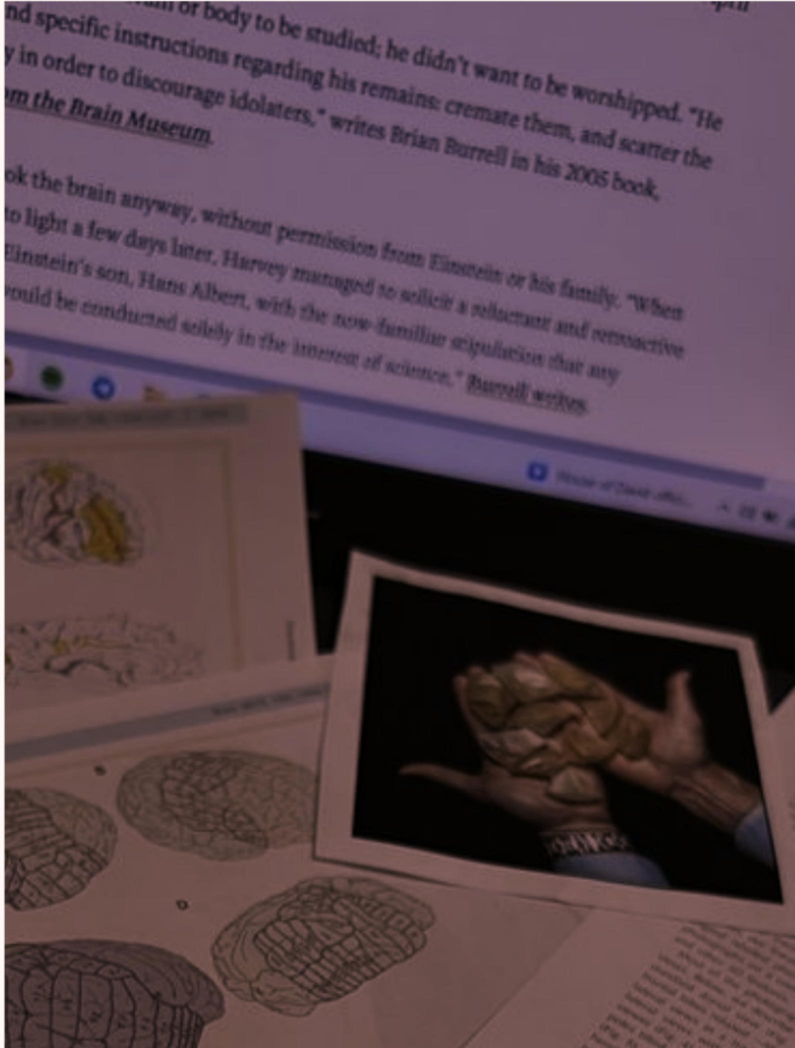

BREATHING EXERCISE

Overstimulated? Scan this QR-code to be sent to a breathing exercise video to reduce stress on both your mind and body.

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NeuroSphere is an international youth-led organization dedicated to raising awareness about neurodegenerative diseases while mentoring and inspiring the next generation of neuroscience leaders.

EXPLORING THE BRAIN, ONE ISSUE AT A TIME

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